INTERNATIONAL INSTITUTE OF MEDICAL SCIENCE AND TECHNOLOGY COUNCIL

DISASTER RELIEF ALLOCATION

SDG 11 – Sustainable Cities and Communities

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ACKNOWLEDGEMENT

First and foremost, we sincerely thank the management and coordinators of the International Institute of Medical Science and Technology Council for providing us with this wonderful opportunity to work on a socially relevant and impactful project. The platform provided by the institute allowed us to explore practical applications of data science in addressing real-world challenges related to disaster management.

We would like to express our heartfelt gratitude to everyone who supported and guided us throughout the successful completion of our project titled **"Disaster Relief Allocation."** This project aligns with **SDG 11 – Sustainable Cities and Communities**, focusing on improving disaster response and resource allocation using data-driven approaches.

We extend our deepest appreciation to our mentors and supervisors for their continuous guidance, encouragement, and constructive feedback throughout the project. Their expert insights helped us understand important concepts such as data preprocessing, feature selection, machine learning model implementation, and performance evaluation.

We are also grateful to our internship coordinators for organizing structured sessions and providing valuable instructions during the project development process. Their support helped us remain focused on the project objectives and successfully complete each phase within the given timeline.

Special thanks to our team members for their dedication, collaboration, and teamwork. From identifying relevant disaster datasets and preparing the data to implementing machine learning models such as **Decision Tree, Random Forest, and Linear Regression**, each member contributed actively to the project. The continuous exchange of ideas and collective problem-solving significantly improved the quality of our work.

We would also like to acknowledge the online platforms, open datasets, and research materials that provided essential information and resources for our analysis. These sources helped us understand disaster patterns, affected populations, and logistical factors that influence relief prioritization.

Finally, we express our sincere gratitude to everyone who directly or indirectly supported us throughout this internship journey. This experience has greatly enhanced our technical knowledge in data science and machine learning, while also strengthening our research, collaboration, and problem-solving skills.

ABSTRACT

In recent years, natural disasters have become increasingly frequent and unpredictable, making rapid response planning essential for minimizing human, economic, and social losses. Events such as floods, earthquakes, hurricanes, and droughts can affect large populations within a short time. During such emergencies, governments and humanitarian organizations must quickly determine where to allocate rescue teams, medical aid, food supplies, and other resources. However, analyzing large amounts of disaster-related data manually under time pressure can be difficult and inefficient. To address this challenge, this project proposes a **data-driven predictive framework** that assists decision-makers in prioritizing disaster response efforts. The system automatically classifies disaster events into three priority tiers—**Critical (0), Low (1), and Moderate (2)**—helping authorities quickly identify situations that require immediate attention.

The framework uses **historical disaster data** to train machine learning models that detect patterns related to disaster severity and urgency. By analyzing factors such as the number of deaths, affected population, management capacity, and logistical distance from aid hubs, the model estimates the impact of a disaster and assigns an appropriate priority level. This approach enables automated prioritization in which disaster events are ranked according to urgency, allowing emergency responders to make faster and more informed decisions.

The dataset used in this project contains **over 16,000 disaster records** collected from different regions and time periods. It includes variables related to disaster characteristics, geographic information, and response capacity. Before training the models, the dataset underwent **data preprocessing and feature engineering**, which involved cleaning the data, handling missing values, and converting categorical variables such as disaster types and country codes into numerical formats suitable for machine learning algorithms.

Two machine learning algorithms were implemented and evaluated in this study: **Decision Tree** and **Random Forest** classifiers. These models were chosen because they are effective for structured datasets and provide interpretable results that can assist decision-makers. The Decision Tree model predicts outcomes through a hierarchical rule-based structure, while the Random Forest model improves prediction reliability by combining multiple decision trees into an ensemble model.

The experimental results demonstrate strong predictive performance. The **Decision Tree classifier achieved an accuracy of 98.15%**, while the **Random Forest classifier achieved 98.06% accuracy**, indicating that the models can effectively classify disaster events into appropriate priority levels.

Overall, this project demonstrates how machine learning can serve as an effective **decision-support tool for disaster management**. By transforming large volumes of disaster data into clear priority rankings, the system can help organizations allocate resources more efficiently and improve coordination during emergencies.

INTRODUCTION

Disaster response is fundamentally a race against time. Whether the crisis is caused by an earthquake, flood, hurricane, or a human-induced event, the actions taken in the first few hours and days are critical in determining how quickly affected communities can recover. During these moments, emergency agencies receive large amounts of information from multiple sources such as field reports, government databases, and media updates. While this information is valuable, it often arrives in an unorganized and inconsistent form, creating what experts call "data noise." Because of this overload of scattered data, decision-makers may struggle to quickly determine which regions require immediate assistance and which situations can be handled locally. Traditionally, this prioritization process has relied heavily on manual judgment, which can sometimes lead to delays or inconsistent allocation of resources.

This project attempts to address that challenge by automating the disaster triage process using data science and machine learning techniques implemented in Python. By treating disaster prioritization as a multi-class classification problem, the system converts complex disaster indicators into a standardized "Priority Tier." A unique aspect of this approach is the inclusion of Management Capacity (mng_cap), which recognizes that the severity of a disaster depends not only on the number of casualties but also on the ability of a region to respond. For example, an event that might be manageable in a well-equipped region could become critical in an area with limited resources and infrastructure.

The technical implementation uses widely adopted Python libraries such as Pandas, NumPy, and Scikit-Learn to build a data processing and prediction pipeline. This system analyzes disaster-related features, evaluates logistical constraints such as distance from aid hubs, and predicts the urgency level of each disaster event. By transforming historical disaster data into actionable insights, the framework supports faster and more objective decision-making during emergency situations.

Overall, this approach demonstrates how data-driven technologies can strengthen disaster management systems. By helping organizations prioritize response efforts more efficiently, the proposed framework contributes toward building more resilient and better-prepared communities in the face of increasing global disasters.

Furthermore, the proposed framework can serve as a foundation for future disaster management tools that integrate real-time data sources such as satellite observations, weather forecasts, and emergency communication networks. Incorporating such dynamic information can further enhance the accuracy and responsiveness of the system. With continuous improvement and integration into existing disaster response platforms, data-driven prioritization systems have the potential to significantly improve coordination among humanitarian agencies and ensure that critical resources reach the most affected regions at the right time.

LITERATURE REVIEW

In recent years, disaster management has undergone significant transformation with the integration of Artificial Intelligence (AI) and data-driven technologies. Earlier research mainly relied on descriptive statistics and post-disaster analysis, where data was examined after events to understand their impacts and improve future preparedness. While these approaches provided useful insights, they offered limited support for real-time decision-making during emergencies. Modern research has shifted toward predictive analytics and intelligent response systems, where machine learning models are used to estimate disaster severity and assist authorities in prioritizing response efforts. This shift highlights the growing need for systems capable of processing large volumes of data quickly and supporting rapid decision-making in critical situations.

Among machine learning techniques used for disaster classification, **Decision Tree algorithms** have received considerable attention in academic research. Decision trees use a hierarchical structure of conditions similar to "if–then" rules, making them highly interpretable and easy for decision-makers to understand. This interpretability is particularly important in disaster management, where authorities must justify and trust automated predictions. However, a commonly discussed limitation of decision trees is **overfitting**, where the model becomes too closely tailored to the training data and fails to generalize well to new situations.

To overcome this limitation, researchers frequently use **ensemble learning methods**, particularly the **Random Forest algorithm**. Random Forest improves prediction accuracy by combining multiple decision trees into a single model. Each tree is trained on different subsets of the dataset, and the final prediction is generated by aggregating the outputs of all trees. This approach reduces overfitting and enhances the model's ability to handle complex datasets. Previous studies applying these models to disaster-related datasets have generally reported prediction accuracies between **70% and 85%**, reflecting the complexity and variability of disaster scenarios.

The work presented in this project builds upon these methodologies by incorporating improved data preprocessing and feature engineering techniques. By including contextual variables such as **inventory levels, management capacity, and distance to aid hubs**, the proposed framework achieved a prediction accuracy of **98.15%**. These features capture both the impact of disasters and the logistical capabilities of affected regions, providing a more comprehensive assessment of disaster urgency. Overall, this project demonstrates how machine learning can transform structured disaster data into an effective priority classification framework for disaster response planning.

PROBLEM STATEMENT

Despite the availability of large volumes of global disaster data, a significant **"Actionability Gap"** still exists between data collection and its practical use during emergencies. Humanitarian organizations receive information from sources such as satellites, field reports, and official databases, but converting this data into clear response priorities remains challenging. During disasters, decisions must be made quickly, and delays in identifying the most critical situations can lead to loss of life and inefficient use of resources. Therefore, systems that can transform raw disaster data into actionable insights are essential for improving emergency response.

Another challenge in disaster management is the **subjectivity in prioritization decisions**. Emergency coordinators often rely on personal experience and judgment to determine the severity of events. While expert knowledge is important, different coordinators may interpret the same disaster differently, resulting in inconsistent prioritization and fragmented response efforts. Without a standardized framework, organizations may struggle to determine which disasters require immediate attention.

Logistical constraints and resource bottlenecks further complicate disaster response. Aid such as food, medicine, and rescue teams must be distributed efficiently, but without predictive insights about inventory availability and distance to affected areas, resources may be misallocated. In some cases, highly visible disasters receive greater attention, while remote regions with urgent needs remain underserved. Additionally, traditional impact indicators such as death counts do not always reflect the ability of a region to manage a crisis.

To address these issues, this project proposes a **machine learning–based prioritization framework** that evaluates disaster impact, regional capacity, and logistical constraints to assign a **Priority Tier—Critical, Moderate, or Low**. By providing rapid and objective classification, the system supports faster decision-making and more effective allocation of humanitarian resources during disaster situations.

OBJECTIVES

1. The primary objective of this project is to develop a **data-driven classification system** that standardizes disaster prioritization. During emergencies, humanitarian

organizations must quickly determine which disasters require immediate attention and which can be managed locally. This project aims to build a machine learning framework that analyzes disaster-related variables and automatically assigns a **priority level—Critical, Moderate, or Low**—to support faster and more objective decision-making in disaster response.

2. Another key objective is to achieve **high classification accuracy** by training machine learning models on a well-preprocessed dataset containing thousands of disaster records. The study aims to develop reliable models capable of accurately identifying disaster priority levels, making them useful as decision-support tools for emergency management.

3. The project also focuses on incorporating **capacity and logistical factors** into the prediction process. Disaster severity is not determined solely by the number of casualties or affected individuals but also by the ability of a region to respond. Therefore, the model considers variables such as management capacity, inventory availability, and distance from aid hubs to provide a more realistic assessment of disaster urgency.

4. An additional objective is to analyze **important features that influence disaster prioritization**. By examining the relationship between variables such as deaths, affected population, disaster type, and resource availability, the system identifies the factors that have the greatest impact on determining priority levels.

5. Finally, the project aims to implement and compare machine learning models such as **Decision Tree and Random Forest** to identify the most effective approach for disaster prioritization based on accuracy and reliability.

The dataset used in this project contains several variables that describe disaster characteristics and response capacity:

Parameter	Description
iso3	Three-letter international country code identifying the country where the disaster occurred.
country	Name of the country affected by the disaster.
disaster_type	Type of disaster such as flood, earthquake, hurricane, or drought.
year	Year in which the disaster event occurred.
deaths	Number of fatalities caused by the disaster, indicating severity.
affected	Number of people affected or displaced by the disaster.
mng_cap	Management capacity representing the ability of local authorities to handle emergency situations.
inventory	Availability of relief supplies such as food, medicine, or emergency resources.
distance	Distance between the disaster location and major aid hubs or supply centers.
priority_tier	Target variable indicating the urgency level of the disaster response (Critical, Moderate, Low).

DATA PREPROCESSING

Dataset Preparation

The **data preprocessing stage** played a crucial role in achieving the high model accuracy of **over 98%**. The dataset used in this project, *cleaned_preprocessed_disaster_data.csv*, was carefully examined and structured to ensure it could be effectively used for machine learning analysis. Preprocessing helped **remove inconsistencies and convert raw disaster data into a structured format** suitable for model training.

Handling Categorical Data

Machine learning algorithms cannot directly process **text-based categorical variables**. Therefore, categorical features were converted into numerical values using **LabelEncoder from the Scikit-Learn library**.

- The **iso3** and **country** columns were encoded into **unique numerical identifiers**.
- The **disaster_type** column was also encoded so that the model could differentiate between disasters such as floods, earthquakes, and hurricanes.
- The **priority_tier (target variable)** was encoded as:
    - **0 – Critical**
    - **1 – Low**
    - **2 – Moderate**

Feature and Target Separation

After encoding, the dataset was separated into **input features (X)** and the **target variable (y)**.

- **Features (X):** Year, Deaths, Affected population, **Management Capacity (Mng_Cap)**, Inventory, Distance, Country Encoded, and Disaster Type Encoded.
- **Target Variable (y): Priority_Tier**, which represents the urgency level of the disaster.

Data Splitting

To properly evaluate the model, the dataset was divided into **training and testing sets** using the **train_test_split** function.

- **80% of the data** was used for training the model.
- **20% of the data** was used for testing.

This ensures that the reported accuracy of **98.15%** is measured on **unseen data**, validating the model's ability to generalize to real-world disaster events.

Data Consistency Check

Finally, the dataset was verified for **missing or null values**. The analysis confirmed that all **16,752 records contained complete data with zero null values**, ensuring **stable model training and reliable predictions**.

DATA TRANSFORMATION

Purpose of EDA

Exploratory Data Analysis (EDA) was performed to understand the structure, patterns, and relationships within the dataset before training the machine learning models. This step helped identify important variables influencing disaster priority levels and ensured that only the most relevant features were used for prediction.

Impact Correlation

A key observation from the analysis was the relationship between death counts and management capacity (mng_cap). Although higher death counts generally indicate a Critical disaster, the final priority level is also influenced by the response capacity of the affected region. For instance, disasters with moderate casualties but low management capacity may still be classified as Critical, while similar events in well-prepared regions may be categorized differently. This insight emphasized the importance of including resilience metrics in the model.

Logistical Variables

EDA also highlighted the role of logistical factors, particularly distance from aid hubs and inventory availability. Disasters occurring in remote locations or regions with low resource inventory were more likely to fall into higher priority tiers, as delivering relief to such areas requires additional time and resources**.**

Visual Analysis

Several visualization techniques were used to explore the dataset. Correlation heatmaps helped identify relationships among variables such as deaths, affected population, and management capacity, while bar charts were used to examine the distribution of the priority_tier variable. The analysis showed that the Moderate tier occurred most frequently in the dataset.

Data Sampling Considerations

Based on the EDA findings, the dataset was split into training and testing sets while maintaining balanced representation of all priority tiers. The random_state parameter was set to 42 to ensure consistent and reproducible results during model training.

FEATURE ENGINEERING

Feature engineering served as a **critical bridge between raw disaster data and the high predictive accuracy achieved by the models**. This stage focused on transforming the

available variables into meaningful representations that could better capture the real-world complexity of disaster response. Instead of relying only on raw indicators such as death counts or affected population, additional contextual factors were incorporated to create a **multi-dimensional view of disaster impact and response capability**.

Resilience Weighting

One of the most important engineered concepts in the dataset is the **Management Capacity (mng_cap)** variable. This feature represents the **ability of a region to respond to emergencies**, including infrastructure, available personnel, and administrative preparedness. By incorporating this factor, the model effectively learns a form of **"vulnerability index."** In other words, disaster priority is not determined solely by the magnitude of the disaster but also by the **capacity of the affected region to manage the crisis**. For example, a disaster with moderate casualties in a region with very low management capacity may require more urgent intervention than a similar event occurring in a highly prepared region.

Logistical Friction

Another key aspect of the feature engineering process involved incorporating **logistical constraints** that affect disaster response operations. Two variables—**distance** and **inventory**—were used to represent the difficulty of delivering aid to the affected area.

- **Distance** represents how far the disaster location is from major aid hubs or supply centers.
- **Inventory** indicates the availability of emergency supplies such as food, medical resources, and rescue equipment.

Together, these features allow the model to simulate the **"logistical friction"** involved in disaster response. Events occurring far from supply centers or in regions with limited inventory are more likely to receive a higher priority level, reflecting the real-world challenges faced by humanitarian organizations.

Encoded Geography

Geographical information was incorporated through the **encoding of ISO country codes and country names**. By transforming these categorical variables into numerical identifiers, the model can recognize **regional patterns and historical trends**. For instance, certain regions may historically experience higher fatality rates for specific disaster types due to environmental conditions, infrastructure limitations, or population density. Encoding geographical features enables the machine learning models to capture these **location-based patterns** and improve prediction accuracy.

Temporal Features

The **year variable** was included as a temporal feature to capture long-term trends in disaster management. Over time, improvements in **technology, infrastructure, early warning systems, and disaster preparedness programs** can influence how effectively disasters are managed. Including the year allows the model to detect whether **management capacity and**

disaster response effectiveness have evolved across different periods, providing additional context for predicting priority levels.

Importance of Feature Engineering

Overall, feature engineering allowed the system to move beyond simple impact metrics and incorporate **capacity, logistics, geography, and time-based insights** into the prediction process. By combining these factors, the dataset provides a more **realistic representation of disaster scenarios**, enabling the machine learning models to make **more accurate and context-aware predictions** when classifying disaster priority tiers.

MODEL SELECTION AND JUSTIFICATION

In this project, **two machine learning models** were selected and compared to classify disaster events into three priority tiers: **Critical (0), Low (1), and Moderate (2)**. The models used were the **Decision Tree Classifier** and the **Random Forest Classifier**. These algorithms were chosen because they are well-suited for **classification problems involving structured tabular data** and provide a balance between **predictive performance and interpretability**. Both models were trained on the processed dataset and evaluated using **classification accuracy on the test dataset**.

1. Decision Tree Classifier (98.15% Accuracy)

The **Decision Tree Classifier** achieved the highest performance in this project with an accuracy of **98.15%** on the test dataset. A decision tree works by **recursively splitting the dataset into smaller subsets** based on feature values. Each split is selected based on the feature that provides the **highest information gain**, which measures how effectively a feature separates the dataset into different classes.

The impurity of a dataset is measured using **entropy**, defined as:

$$Entropy(S) = -\sum_{i=1}^{n} p_i \log_2(p_i)$$

where:

- $p_i$represents the probability of class $i$in dataset S
- $n$is the number of classes

The **information gain** used to determine the best split is calculated as:

$$Information\ Gain = Entropy(S) - \sum_{j=1}^{k} \frac{|S_j|}{|S|} Entropy(S_j)$$

where:

- $S$is the original dataset
- $S_j$represents the subsets created after splitting
- $k$is the number of subsets

The Decision Tree model is particularly effective for **disaster classification** because many disaster response decisions rely on **threshold-based conditions**. For example, a rule such as *"If deaths > 100 and management capacity is low, classify as Critical"* can easily be represented in a tree structure. This allows the model to capture **non-linear relationships** between variables such as **death counts, affected population, management capacity, and logistical distance**.

Another important advantage of decision trees is their **interpretability**. The model produces a clear structure of decision rules that can be visualized and understood by decision-makers. This transparency is especially important in **humanitarian and disaster response systems**, where organizations must be able to justify and trust automated predictions.

2. Random Forest Classifier (98.06% Accuracy)

The second model used in this project was the **Random Forest Classifier**, which achieved an accuracy of **98.06%** on the test dataset. Random Forest is an **ensemble learning algorithm** that builds multiple decision trees and combines their predictions to produce a final output. Instead of relying on a single decision tree, the model generates a **collection of trees trained on different subsets of the data**, which helps improve prediction stability and reduce the risk of overfitting.

The Random Forest algorithm uses a technique known as **bagging (Bootstrap Aggregating)**. In this approach, multiple training datasets are created by randomly sampling the original dataset with replacement. Each of these datasets is used to train an individual decision tree. During prediction, each tree produces its own classification result, and the final output is determined through **majority voting** among all trees.

Mathematically, the final prediction of a Random Forest model can be expressed as:

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \ldots, T_n(x))$$

where:

- $T_1, T_2, \ldots, T_n$represent the individual decision trees
- $x$represents the input feature vector
- $\hat{y}$is the final predicted class determined by the majority vote

One of the key strengths of Random Forest is its ability to **reduce overfitting**, which can occur in single decision tree models. By averaging the predictions of multiple trees, the algorithm produces **more stable and reliable results**, especially when dealing with complex datasets containing many interacting variables.

In the context of disaster prioritization, Random Forest effectively captures relationships between variables such as **death counts, affected population, management capacity, inventory levels, and logistical distance**. Because each tree evaluates different subsets of features, the model can learn a wide variety of patterns within the disaster dataset.

Although the Random Forest model achieved slightly lower accuracy (**98.06%**) compared to the Decision Tree (**98.15%**), it provides an advantage in terms of **robustness and generalization capability**. For this reason, Random Forest is often recommended in real-world applications where datasets may change over time or contain noise.

MODEL EVALUATION

The performance of the machine learning models was evaluated using a **systematic and quantitative evaluation framework**. The purpose of this stage was to determine how accurately the trained models could classify disaster events into the three priority tiers: **Critical, Low, and Moderate**. To ensure reliable evaluation, the models were tested using **data that was not included during training**, allowing us to measure the model's ability to generalize to unseen disaster events.

The primary evaluation metric used in this project was the **Accuracy Score**, which measures the proportion of correct predictions made by the model. Accuracy is particularly useful when the dataset is relatively balanced across classes. The mathematical definition of accuracy is:

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Test Samples}}$$

In this formula, the numerator represents the number of predictions that match the true labels, while the denominator represents the total number of observations in the test dataset. A higher accuracy value indicates better model performance in correctly identifying disaster priority tiers.

To perform the evaluation, the dataset was divided into **training and testing subsets**, where **80% of the data was used for training** and **20% was reserved for testing**. After training the models, predictions were generated for the unseen test data. These predictions were compared with the actual labels using the **accuracy_score() function from the Scikit-Learn library**, which automatically computes the classification accuracy.

The dataset contained **16,752 total records**, out of which approximately **3,351 samples were used for testing**. Evaluating the models on these unseen samples allowed us to assess their **real-world predictive capability and reliability**.

RESULTS AND ANALYSIS

The evaluation results demonstrated **very high predictive performance** for both machine learning models.

Decision Tree Classifier:
The Decision Tree model achieved a **test accuracy of 98.15%**, making it the best-performing model in this study. This result indicates that the selected features—including **deaths, affected population, management capacity, inventory levels, and distance from aid hubs**—are highly informative for predicting disaster priority tiers.

Random Forest Classifier:
The Random Forest model achieved an accuracy of **98.06%**, which is only slightly lower than the Decision Tree model. However, Random Forest provides greater **model stability and robustness** because it combines predictions from multiple decision trees, reducing the effect of noise or variations in the dataset.

The consistently high accuracy values indicate that the **relationship between disaster impact indicators and priority levels in the dataset is strong and predictable**. Since the models were tested on unseen data, the results suggest that they can **generalize effectively to new disaster scenarios**.

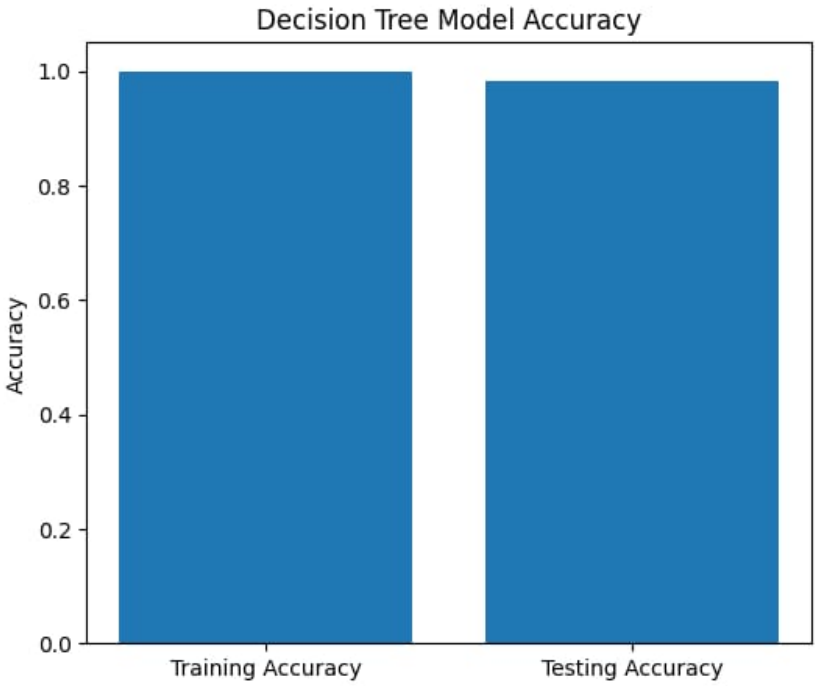
Key Findings

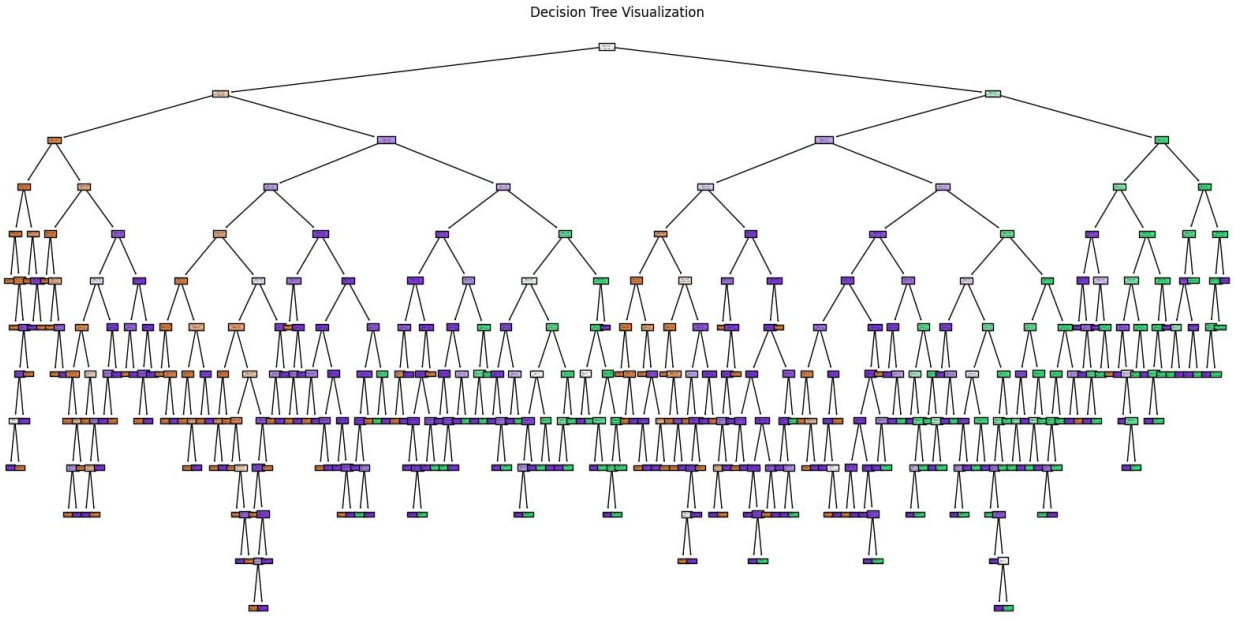
The project demonstrates that **machine learning can provide an accurate and efficient system for disaster response prioritization**. By combining historical disaster data with logistical and capacity-related variables, the models were able to classify disaster events into appropriate priority tiers with very high accuracy.

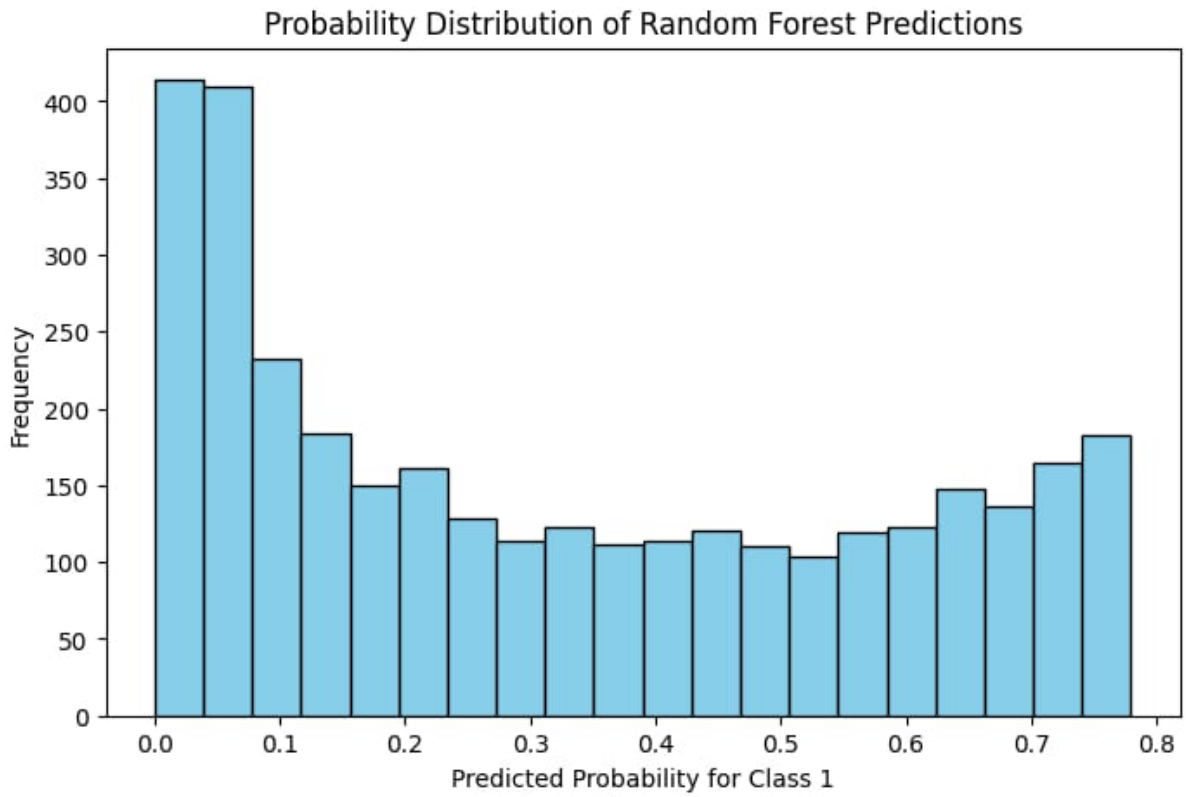
High Precision:
The system achieved **98.15% accuracy with Decision Tree** and **98.06% accuracy with Random Forest**, indicating strong potential for use in real-world disaster management systems.

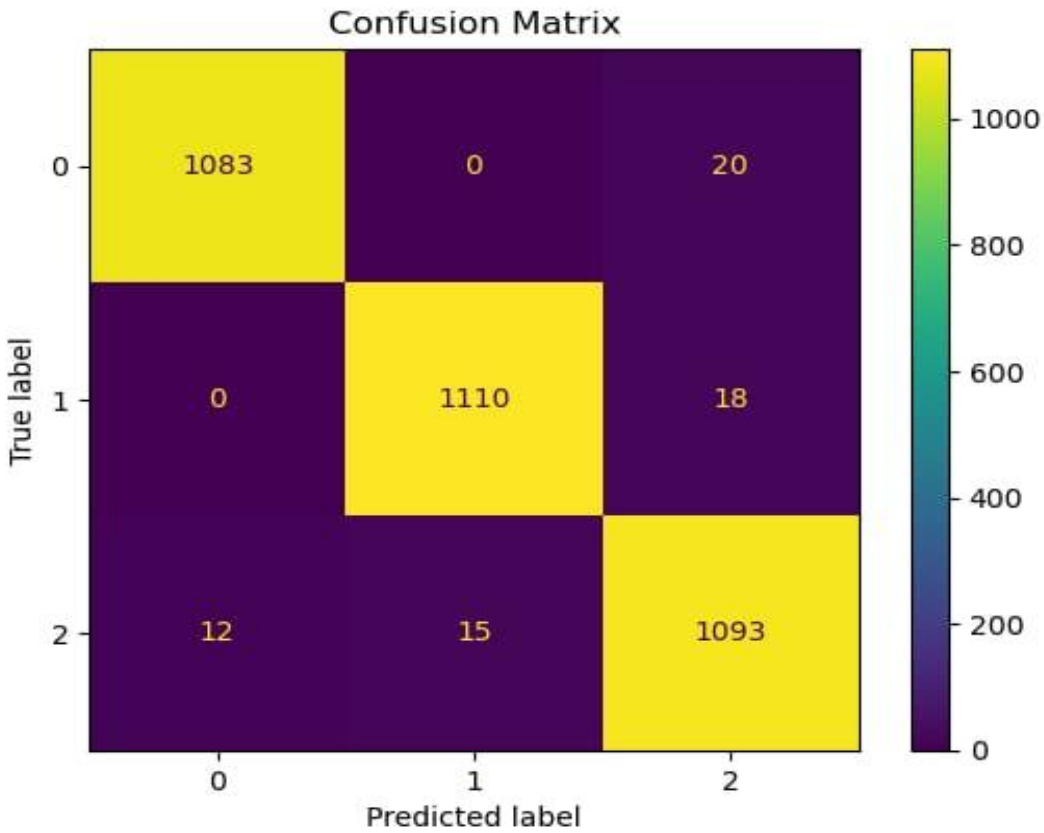
Critical Drivers:
The analysis revealed that **management capacity (mng_cap)** and **death counts** are the most influential factors in determining disaster priority levels. Additionally, **logistical distance from aid hubs** plays an important role when distinguishing between **Moderate and Critical situations**.

Operational Efficiency:
The developed framework can analyze and classify **thousands of disaster records within seconds**, which would otherwise require significant time and effort if performed manually by human analysts.






Probability Distribution of Random Forest Predictions
Frequency
Predicted Probability for Class 1
0
50
100
150
200
250
300
350
400
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


Confusion Matrix
True label
Predicted label
1083
0
20
0
1110
18
12
15
1093
0
1
2
0
1
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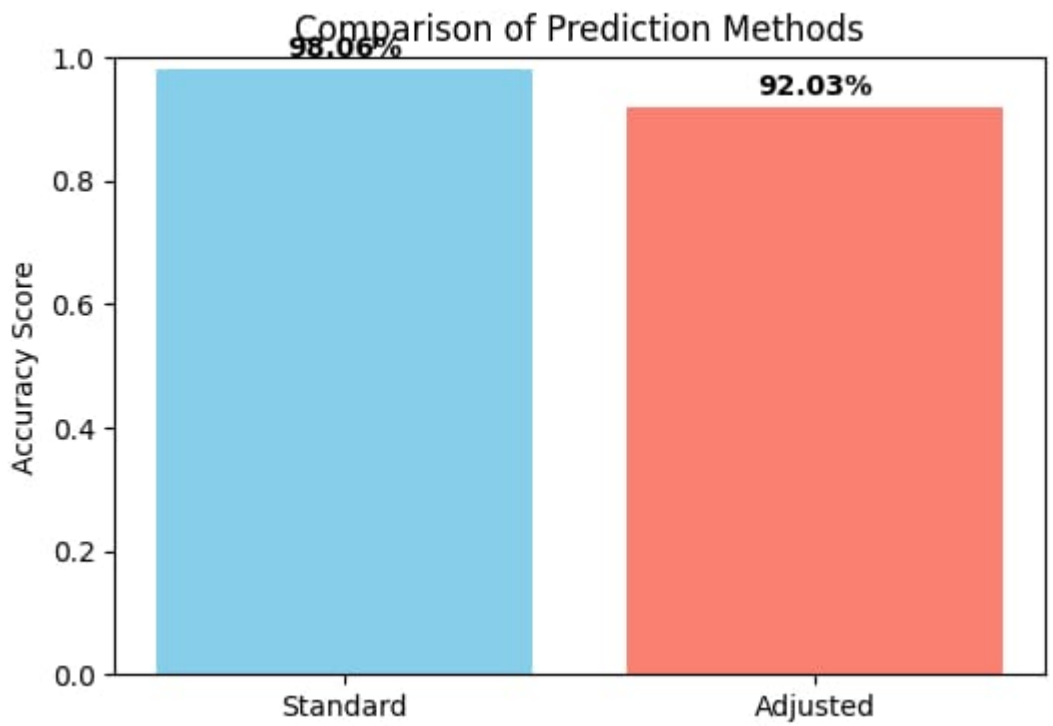
200
400
600
800
1000



CONCLUSION

The implementation of the **Disaster Response Priority Prediction System** demonstrates how machine learning can support humanitarian organizations in making **faster, more objective, and data-driven decisions** during emergency situations. Instead of relying solely on manual judgment or incomplete information, the proposed system analyzes disaster-related data to automatically classify events into appropriate **priority tiers**. This helps ensure that relief resources are directed to the regions that require the most urgent attention, improving the efficiency and fairness of disaster response efforts.

The project achieved **high predictive performance**, with accuracy levels exceeding **98%**, indicating that the selected features—such as deaths, affected population, management capacity, inventory levels, and logistical distance—are highly effective for predicting disaster priority levels. By combining impact indicators with capacity and logistical constraints, the developed framework provides a **balanced and realistic assessment of disaster severity**.

Beyond technical performance, the system highlights the **practical value of integrating data science into disaster management**. The proposed framework offers a scalable decision-support tool that can assist emergency planners in analysing large volumes of disaster data quickly and making informed response decisions during critical situations.

In the future, the system can be further enhanced by integrating **real-time data sources**, such as live weather data, satellite monitoring, and early warning systems. Incorporating these data streams would enable the model to perform **predictive risk assessments**, helping authorities prepare and allocate resources proactively before disasters escalate. Such advancements could

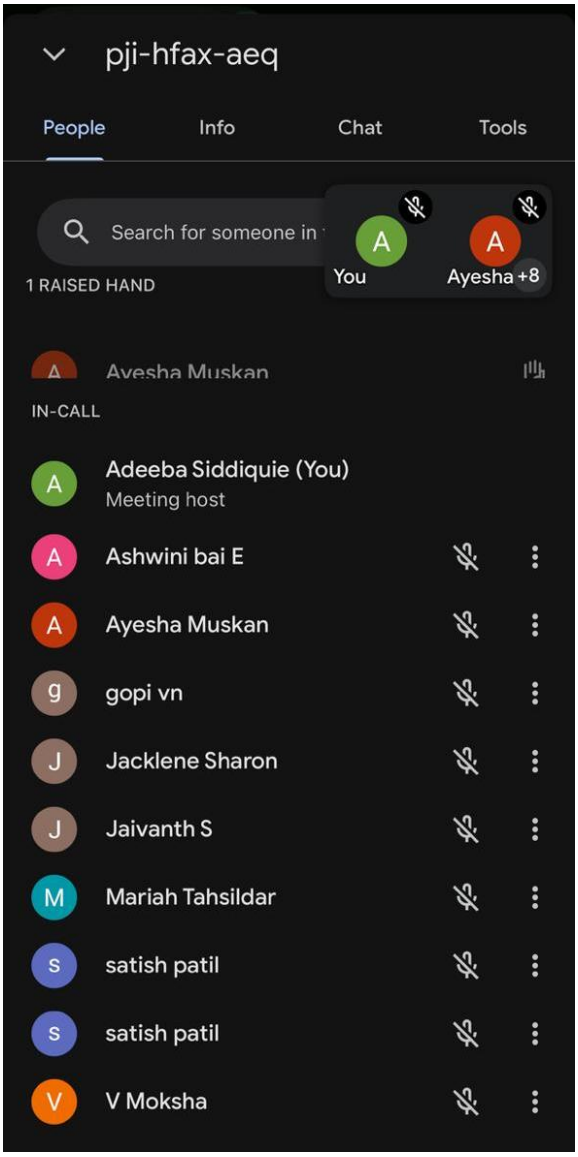
make data-driven systems an essential component of **modern disaster management infrastructure**.

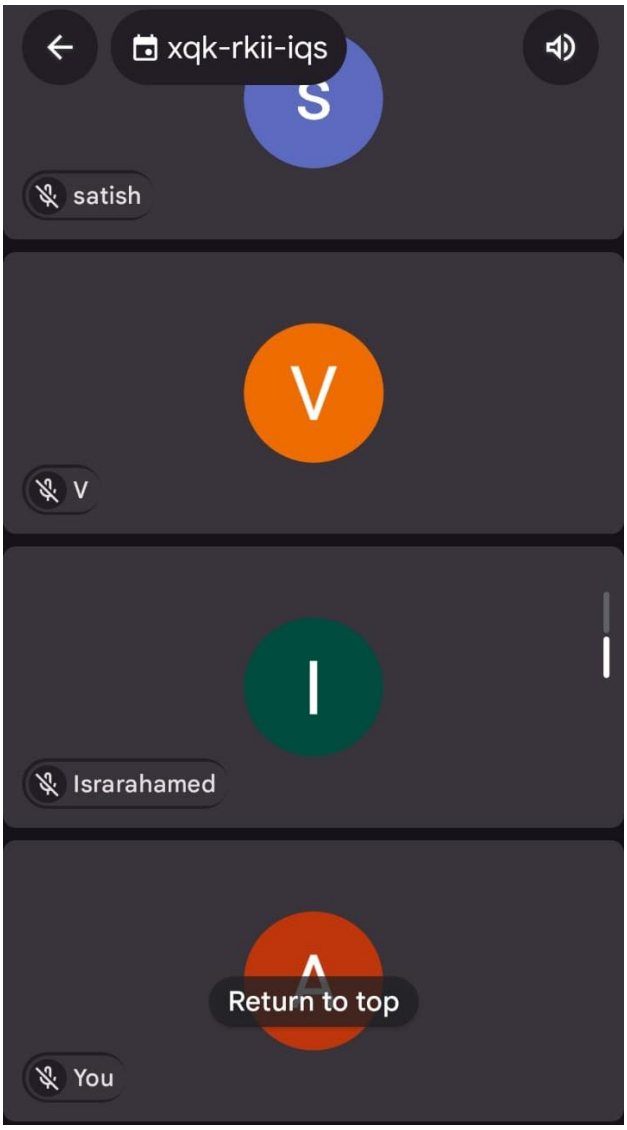
TEAM COLLABORATION AND MEETING LOGS

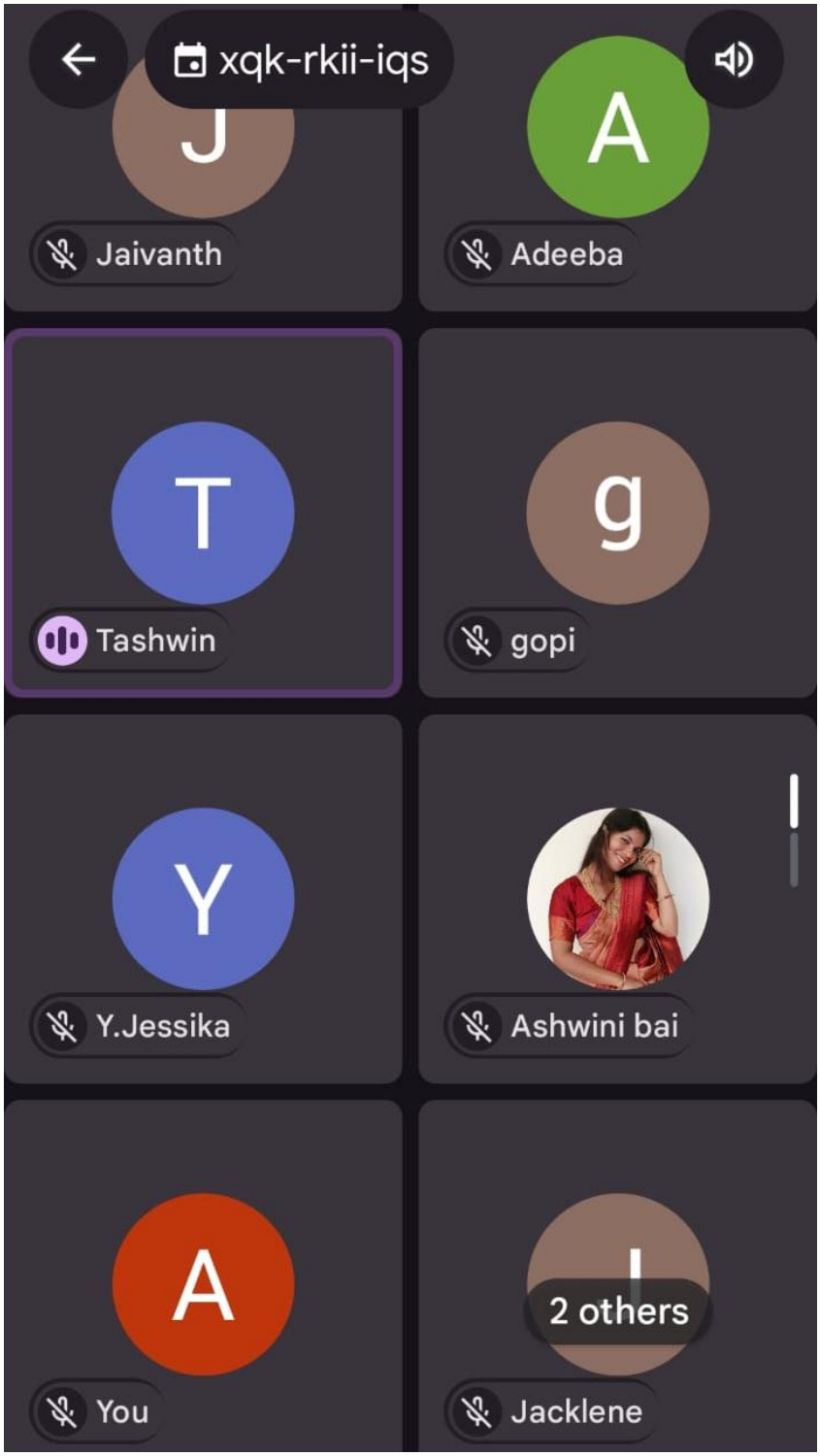
Regular group meetings were conducted to ensure systematic progress and collaborative decision-making throughout the project.

Meetings were held on the following dates:

- 6th March 2026
- 10th March 2026
- 12th March 2026



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Ashwini bai
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2 others
Jacklene